

Causal Evaluation and Stochastic Revealed Preferences with Unobserved Randomness: A Structural Simulation Approach

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May 19, 2023

Abstract. Conventional approaches to causal inference analysis, such as the Difference-in-Differences method (DiD), are susceptible to bias due to unobserved measurement error in the data, thus distorting the inference of the effect from a given shock or intervention over a variable of interest. In this paper, consumption data is simulated to follow a situation in which a subset of a population is randomly assigned an exogenous shock to their demand for a given good in their consumption bundle at some moment of time, while suffering from measurement error. The methodology applied, recovered from Aguiar and Kashaev (2020), provides a framework for testing the causal effect of any such situation in the households' bundle of consumption, leading—given the structural nature of this methodology—to testing the change in the structure of the households' preferences and identifying an Average Treatment Effect on the Treated (ATT) set. In contrast to the bias attained by a conventional DiD reduced-form estimation, the proposed structural estimation accounts for existing unobserved measurement error in the data.

Keywords: Microeconomics; Revealed Preferences; Structural Estimation; Causal Inference; Set-Identification; Difference-in-Differences; Econometric Methodology.

JEL Classification: C14, C15, D12, D81.

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1. Introduction

What is the causal effect of an exogenous shock hitting a population's demand for a given good on their consumption bundle composition? While some works in the development economics literature have tackled the effect of such kind of situations, existent approaches have been mainly based on reduced-form specifications relying on conventional causal inference methods (Angelucci & De Giorgi, 2009; Cunha, De Giorgi, & Jayachandran, 2019; O. Attanasio et al., 2010; Oosterbeek, Ponce, & Schady, 2008).

One standard approach consists in calculating the change in the demand share of a given good in the consumption bundle of a population with respect to full bundle expenditure, via a reduced-form specification. These conventional approaches to causal testing in consumption data are vulnerable to unobserved measurement error, thus biasing the estimated effect and potentially distorting the inferred magnitude of any such effect. This paper presents a framework for estimating causal Average Treatment effects on the Treated (ATT) on a population for the specific case of the change in the consumption bundle of a population facing exogenous shocks. We introduce the structural revealed preference framework, in the fashion of Aguiar and Kashaev (2020), to a parametric Cobb-Douglas utility function. Importantly, unlike a reduced-form estimation, this approach allows for accounting for unobserved randomness in the data (e.g., measurement error) while making results interpretable in terms of a change in the structure of preferences of the population.

A data generating process (DGP) is simulated to portray a population in the following situation: Decision-Makers (DM) maximize their individual utility subject to a budget constraint at each period of time for T periods. At a given moment, a positive random exogenous shock hits the demand of a random subset of this population. This effect on the observed consumption shares for each good of the DM's bundle can be causally estimated by a reduced-form Difference-in-Differences (DiD) approach. However, the DGP is induced to contain unobserved randomness, thus portraying a phenomenon observed in real data (Bound, Brown, & Mathiowetz, 2001). This issue is solved by structurally estimating this causal effect via a system of moment conditions on the structural model that accounts for unobserved randomness in the data. In this case, this takes the form of measurement error, for which a single centering assumption is considered. It is later shown that, while the reduced-form DiD estimation is biased, the structural model allows for the inference of an identified set that asymptotically matches the effect yielded by the reduced-form ATT in the "unobserved" real data without measurement error. In other words, by properly modeling the centered measurement error's moment condition in our model, we can infer an asymptotically unbiased causal estimator despite data suffering from measurement error.

This strategy exploits recent methodological developments in the Revealed Preferences (RP) literature. It aims not only to causally test the change of the observed consumption bundle of a population in the face of an exogenous shock, but to statistically test the preference structure evolution of the treatment group over multiple time periods and for a given menu of consumption goods. Our approach is parametric and provides a structural formulation to estimate the interest parameter, using the moment conditions approach and the robustness of the method provided by Aguiar and Kashaev (2020). This testing scheme is made operational via an algorithm designed to deal with multivariate conditional distributions over latent vectors,

implemented through the Entropic Latent Variable Integration via Simulation (ELVIS) method developed by Schennach (2014) over the space of moment conditions.

The paper proceeds as follows. Section 2 presents the model environment. Section 3 presents the causal inference procedure provided by a conventional Difference-in-Differences method and discusses the introduction of measurement error. Section 4 builds on the elements presented in Section 2 to provide an alternative causal estimation that accounts for measurement error using a structural framework. Section 5 provides simulation results. Section 6 concludes with discussion and directions for future research.

1.1. Relation to the Literature

By combining causal intervention evaluation and stochastic revealed preferences in the presence of measurement error, this study contributes to the economics literature by providing a method to disentangle causal effects that would otherwise be biased. With regard to causal intervention evaluation, this paper relates to the evaluation of policy interventions or exogenous shocks on the consumption bundle composition of households. Relevant contributions include Angelucci and De Giorgi (2009), Cunha et al. (2019), O. Attanasio et al. (2010), and Oosterbeek et al. (2008), who apply mostly reduced-form causal inference techniques.

The second area to which this paper contributes is the set of potential applications provided by stochastic revealed preference (RP) analysis. Recent theoretical developments have focused on measuring the fit of observed individual consumption decisions with the classical Utility Maximization Theory (UMT) and its deviations. These approaches have used structural, non-parametric estimation methods to test rationalizability. Consistency with the classical Exponential Discounting (ED) model for intertemporal consumption has been tested as well.

Intending to test rationalizability non-parametrically in an RP context, Kitamura and Stoye (2018) used a testing method built on a model based on a linear program over ‘budget patches’. Others, such as Aguiar and Kashaev (2020), applied such testing in the context of Afriat inequalities, and Demuynck and Potoms (2022) applied a column-generation approach and tools from convex analysis. Kashaev, Gauthier, and Aguiar (2023) provide a further innovation in which a flexible, stochastic, and dynamic approach to RP can handle serial correlation and cross-section heterogeneity in preferences.

While most of these approaches have been non-parametric, we innovate by providing a practical application of such frameworks in a parametric environment that builds upon the empirical consumption bundle composition of households. This applied approach complements relevant contributions from O. P. Attanasio, Meghir, and Santiago (2012); Todd and Wolpin (2006). Other works, such as Deb, Kitamura, Quah, and Stoye (2023), analyze welfare consequences via preferences for prices. Adams (2020) develop a method to rationalize consumption predictions, advancing the revealed preference literature toward the prediction of rational demands over new budget sets.

Our work builds upon Aguiar and Kashaev (2020), particularly by formulating the testing procedure through moment conditions relying on a model-driven first-order conditions approach. Despite the specific form of the moment conditions, our method outperforms the Difference-in-Differences reduced-form specification

when attempting to find the causal effect of a policy program in the presence of measurement error, because we are able to provide a narrower set that asymptotically approximates the true effect. As long as the model can be expressed in the form of moment conditions,¹ this study provides a procedure to correctly evaluate policy in settings where the data suffers from measurement error and other forms of unobserved randomness. Future work will apply this method to Conditional Cash Transfer data in Mexico and Ecuador, exploiting the work of O. Attanasio and Pastorino (2020); Schady, Araujo, Peña, and López-Calva (2008); Oosterbeek et al. (2008).

2. Model Environment

Consider a given number of households determined by $\mathcal{J} = \{1, \dots, J\}$. For a finite set of goods $\mathcal{L} = \{1, \dots, L\}$, each household $j \in \mathcal{J}$ consumes over a space $X \subseteq \mathbb{R}_{++}^L$ for a number of periods given by $\mathcal{T} = \{1, \dots, T\}$. Let the household's preferences be defined by a utility function $u_{j,t}(c_{j,t})$ where, for each $j \in \mathcal{J}$, $u_t(\cdot)$ is continuous, concave, and locally non-satiated for all $t \in \mathcal{T}$, and $c_t \in X$ is the vector of consumption for each household j across all periods $t \in \mathcal{T}$. Consider the case where preferences can be represented by Cobb-Douglas (CD) preferences:

$$u_t(c_t) = \prod_{l=1}^L c_{l,t}^{\alpha_{l,t}} \quad (1)$$

with the property that $\sum_l \alpha_{l,t} = 1$ at each $t \in \mathcal{T}$, so that each good l contributes a budget share $\alpha_{l,t}$ to utility. Since we are modeling this as a “repeated cross-section”, DMs hold no savings and do not discount future income flows. DMs are assumed to have perfect foresight over their exogenous income. The DM's utility maximization problem leads to the following testable condition of interest:

Definition 1 (CD-Rationalizability). A deterministic array $(p_t, c_t)_{t \in \mathcal{T}}$ is CD-rationalizable if, for Cobb-Douglas preferences as in equation (1), $(c_t)_{t \in \mathcal{T}}$ solves:

$$\max_{c_t \in X} \prod_{l=1}^L c_{l,t}^{\alpha_{l,t}} \quad (2)$$

subject to

$$p'_t c_t = y_t \quad (3)$$

As implied by the parametric assumption on preferences, $u(\cdot)$ is concave, and since equation (3) is convex, there exists a unique solution to the utility maximization problem in Definition 1. The following result characterizes this solution.

Theorem 1. For a Cobb-Douglas model of preferences, the following are equivalent:

¹With a centering assumption.

- (i) The deterministic array $(p_t, c_t)_{t \in \mathcal{T}}$ is CD-rationalizable.
- (ii) There exists a matrix $(\alpha_{l,t})_{l \in \mathcal{L}, t \in \mathcal{T}}$ such that $\sum_l \alpha_{l,t} = 1$, and

$$c_{l,t} = \frac{\alpha_{l,t} y_t}{p_{l,t}}$$

for all $t \in \mathcal{T}$.

The setup thus far does not consider any perturbation in the data. The following subsection introduces measurement error.

2.1. Introducing Measurement Error

Assume income y_t is normalized to one for each DM at each period, and denote a random variable in bold as $\mathbf{x}$. Additionally, denote $\mathbf{x}^*$ as a random variable, data set, or array that does not suffer from any perturbation, i.e., measurement error.

As argued by Aguiar and Kashaev (2020), conventional RP testing methodologies—even in fully non-parametric environments—tend to over-reject rationalizability when unobserved randomness (e.g., measurement error) is not accounted for. A similar argument applies to conventional reduced-form causal estimations. Before defining measurement error, CD-rationalizability must be defined in a fully stochastic context:

Definition 2 (s/CD-Rationalizability). A random array $(\mathbf{p}_t^*, \mathbf{c}_t^*)_{t \in \mathcal{T}}$ is said to be statistically CD-rationalizable for a given structure of Cobb-Douglas preferences, if there exists a random matrix $(\mathbf{a}_{l,t})_{l \in \mathcal{L}, t \in \mathcal{T}}$ ² such that:

- (i) Each entry of $(\mathbf{a}_{l,t})_{l \in \mathcal{L}, t \in \mathcal{T}}$ is a random variable supported on or inside a known set $\mathbf{A} \subseteq [0, 1]^{L \times T}$ and $\sum_l \mathbf{a}_{l,t} = 1$ for every $l \in \mathcal{L}$ and every $t \in \mathcal{T}$;
- (ii) $\mathbf{c}_{l,t}^* = \frac{\mathbf{a}_{l,t}}{\mathbf{p}_{l,t}^*}$ a.s. for all $l \in \mathcal{L}$ and all $t \in \mathcal{T}$;
- (iii) For every $l \in \mathcal{L}$ and $t \in \mathcal{T}$,

$$\mathbb{P}\left(\mathbf{c}_{l,t}^* > 0, \mathbf{c}_{l,t}^* = \frac{\mathbf{a}_{l,t}}{\mathbf{p}_{l,t}^*}\right) = 1.$$

For a collection of random sequences $(\mathbf{p}_t^*, \mathbf{c}_t^*)_{t \in \mathcal{T}}$ to be s/CD-rationalizable is equivalent, given some random positive matrix $(\mathbf{a}_{l,t})$ supported on or inside $\mathbf{A} \subseteq [0, 1]^{L \times T}$, to requiring that $\mathbf{c}_{l,t}^* = \frac{\mathbf{a}_{l,t}}{\mathbf{p}_{l,t}^*}$ a.s. for all $t \in \mathcal{T}$. This is proved in a general, non-parametric manner by Aguiar and Kashaev (2020), who also show that this condition does not guarantee a conclusive determination of whether a data set is “almost” or “exactly” s/CD-rationalizable. This makes the following necessary:

Definition 3 (Approximate s/CD-Rationalizability). We say that $(\mathbf{p}_t^*, \mathbf{c}_t^*)_{t \in \mathcal{T}}$ is approximately consistent with

²In Definition 2, $(\mathbf{a}_{l,t})_{l \in \mathcal{L}, t \in \mathcal{T}}$ stands for the stochastic counterpart of the deterministic matrix $(\alpha_{l,t})_{l \in \mathcal{L}, t \in \mathcal{T}}$ stated in Theorem 1.

s/CD-rationalizability if there exists a positive matrix $(\mathbf{a}_{t,l,r})_{t \in \mathcal{T}, l \in \mathcal{L}}$ supported on or inside $\mathbf{A} \subseteq [0, 1]^{L \times T}$ such that

$$\mathbb{P} \left(\mathbb{I} \left(\mathbf{c}_{l,t}^* = \frac{\mathbf{a}_{t,l,r}}{\mathbf{p}_{l,t}^*} \right) = 1 \right) \xrightarrow{r \rightarrow +\infty} 1 \quad (4)$$

for all $t \in \mathcal{T}$.

Consider any non-systematic measurement error on observed data arising from misrecording issues (e.g., trembling-hand errors). As in Aguiar and Kashaev (2020), we *define* this error, and our framework is consistent with any measurement error type satisfying these characteristics. Following Demuynck and Potoms (2022), we may also conceive of this as “unobserved randomness.”

Let the *measurement error* be an array $\mathbf{w} = (\mathbf{w}_t)_{t \in \mathcal{T}} \in W$ given as the difference between recorded and true data for consumption and prices:

$$\mathbf{w}_t = \begin{bmatrix} \mathbf{w}_t^c \\ \mathbf{w}_t^p \end{bmatrix} \quad (5)$$

with $\mathbf{w}_t^c = \mathbf{c}_t - \mathbf{c}_t^*$ and $\mathbf{w}_t^p = \mathbf{p}_t - \mathbf{p}_t^*$.

A random array $(\mathbf{p}_t, \mathbf{c}_t)_{t \in \mathcal{T}}$ is s/CD-rationalizable if and only if there exists an array $(\mathbf{a}_{l,t}, \mathbf{w}_t)_{t \in \mathcal{T}, l \in \mathcal{L}}$ such that

$$\mathbf{c}_{l,t} - \mathbf{w}_{l,t}^c = \frac{\mathbf{a}_{l,t}}{(\mathbf{p}_{l,t} - \mathbf{w}_{l,t}^p)}; \quad \text{a.s., } \forall t \in \mathcal{T}. \quad (6)$$

In what follows, we assume measurement error hits consumption only, not prices.³ Let $\mathbf{e} = (\mathbf{a}', \mathbf{w}') \in E|X$ be the vector of unobserved random latent variables, supported on or inside the conditional support $E|X$, where $\mathbf{x} = (\mathbf{p}_t, \mathbf{c}_t) \in X$ is the observed data. Define $g_M : X \times E|X \rightarrow \mathbb{R}^{d_M}$ as a *measurement error moment*. The only imposed assumption is:

Assumption 1 (Centered Measurement Error). (i) The random vector of latent variables $\mathbf{e}$ is supported on or inside $E|X$, and (ii) there exists a known mapping $g_M : X \times E|X \rightarrow \mathbb{R}^{d_M}$ such that

$$\mathbb{E}[g_M(\mathbf{x}, \mathbf{e})] = 0. \quad (7)$$

3. The Difference-in-Differences Approach

Difference-in-Differences (DiD) is a statistical method to identify causal effects in quasi-experimental frameworks. In its simplest form, the method exploits the random assignment of individuals into treated and untreated groups to compare mean outcomes across groups. The simple DiD method requires two assumptions to identify causal effects: the parallel trends and the no-anticipation assumptions. The former requires

³This is consistent with Appendix B.

that, absent the treatment, the average outcomes of the treated and control groups would have followed a similar trend over time. The latter states that the treatment has no causal effect prior to its implementation (Roth, Sant'Anna, Bilinski, & Poe, 2023).

In the context of an intervention, the econometrician observes an outcome variable Y_{jt} for each household and the group assignment $D_j \in \{0, 1\}$. Assuming that both parallel trends and no-anticipation hold, the Average Treatment effect on the Treated (ATT) is:

$$ATT = \mathbb{E}[Y_{j,t^*} - Y_{j,t^*-1} | D_j = 1] - \mathbb{E}[Y_{j,t^*} - Y_{j,t^*-1} | D_j = 0] \quad (8)$$

where $D_j = 1$ denotes that household j belongs to the treatment group, t^* represents a post-treatment period, and $t^* - 1$ a pre-treatment period.

Given the Cobb-Douglas structure described in Section 2, the econometrician can test whether an intervention changes the consumption share of good l in period t for the treated households. First, consumption data are transformed into shares (see equation (9)). The effect of interest is described by equation (10), where $j \in \tau \equiv D_j = 1$ and $j \in \mathcal{C} \equiv D_j = 0$ denote treatment and control assignment.⁴

$$\mathbf{a}_{j,l,t} = \frac{\mathbf{c}_{j,l,t}^*}{\sum_{k=1}^L \mathbf{c}_{j,k,t}^* \mathbf{p}_{k,t}} \quad (9)$$

$$ATT = \beta = [\bar{\mathbf{a}}_{j \in \tau, t^*} - \bar{\mathbf{a}}_{j \in \tau, t^*-1}] - [\bar{\mathbf{a}}_{j \in \mathcal{C}, t^*} - \bar{\mathbf{a}}_{j \in \mathcal{C}, t^*-1}] \quad (10)$$

The classic method to estimate the ATT uses a two-way fixed effects model where the parameter of interest captures the time-group interaction. The following definition tailors this method to our consumption data model:

Definition 4 (Difference-in-Differences Estimation). Assume that a subset of the population $\tau \subset \mathcal{J}$ receives an exogenous shock effective on their demand for good $l \in \mathcal{L}$. The Difference-in-Differences effect is given by estimating parameter β in:

$$\mathbf{a}_{j,l,t} = \phi_0 + \phi_1 \text{Time}_{j,t} + \phi_2 \text{Group}_{j,t} + \beta (\text{Time}_{j,t} \times \text{Group}_{j,t}) + \epsilon_{j,t} \quad (11)$$

where $\text{Time}_{j,t}$ is a dummy indicating the post-treatment period t^* , $\text{Group}_{j,t}$ is a dummy indicating that j belongs to the treatment group, and $\epsilon_{j,t}$ is the residual term.

In the presence of measurement error in $\mathbf{a}_{j,l,t}$,⁵ given by

$$\tilde{\mathbf{a}}_{j,l,t} = \frac{\mathbf{c}_{j,l,t}^* + \mathbf{w}_{j,l,t}^c}{\sum_{k=1}^L (\mathbf{c}_{j,k,t}^* + \mathbf{w}_{j,k,t}^c) \mathbf{p}_{k,t}},$$

⁴In this notation, $\bar{x}$ stands for the mean of a variable x_i .

⁵Recall that l is held fixed.

with observed consumption $\mathbf{c}_{j,l,t} = \mathbf{c}_{j,l,t}^* + \mathbf{w}_{j,l,t}^c$, the econometrician will estimate a different specification:

$$\tilde{\mathbf{a}}_{j,l,t} = \frac{\mathbf{c}_{j,l,t}}{\sum_{k=1}^L \mathbf{c}_{j,k,t} \mathbf{p}_{k,t}} = \tilde{\phi}_0 + \tilde{\phi}_1 \text{Time}_{j,t} + \tilde{\phi}_2 \text{Group}_{j,t} + \tilde{\beta} (\text{Time}_{j,t} \times \text{Group}_{j,t}) + \epsilon_{j,t}. \quad (12)$$

Although $\mathbb{E}[w_{j,l,t}^c] = 0$, if $\sigma_{w^c} > 0$ is sufficiently large, estimates retrieved by standard DiD methodology could appear non-significant. To the best of our knowledge, there are no statistical reasons to assume that classical measurement error in elicited consumption data is bounded by arbitrary limits. In empirical applications such as surveys, this can cause the inability to reject the null hypothesis, thus erroneously inferring that an exogenous shock did not affect an outcome of interest when it actually did.

Table 1 in Section 5 reports the estimated coefficients for equations (11) and (12), as well as their corresponding confidence intervals. Results are reported as the mean value over 1,000 simulation repetitions. A significance non-rejection rate is also presented in Table 1, indicating the share of repetitions for which the statistical significance at the 95% level is not rejected. A “misinference rate” is also reported, indicating the share of repetitions for which $\hat{\beta}$ is not included in the 95% confidence interval of $\hat{\beta}$.

4. A Framework for Structural Causal Inference

Given the model and first-order conditions stated previously, we apply the testing framework of Aguiar and Kashaev (2020). For the established preference structure, the moment conditions, for each $l \in \mathcal{L}$ and all $t \in \mathcal{T}$, are:

$$g_{A,j,l,t}(\mathbf{x}, \mathbf{e}) = \mathbb{I} \left[\mathbf{c}_{j,l,t} - \mathbf{w}_{j,l,t}^c = \frac{\mathbf{a}_{j,l,t}}{\mathbf{p}_{j,l,t}} \right] - 1 \quad (13)$$

$$g_{D,j,l=1}(\mathbf{e}; \theta) = \left[[\mathbf{a}_{j \in \tau, t^*} - \mathbf{a}_{j \in \tau, t^*-1}] - [\mathbf{a}_{j \in \mathcal{C}, t^*} - \mathbf{a}_{j \in \mathcal{C}, t^*-1}] \right] - \theta \quad (14)$$

$$g_{M,j,l,t}(\mathbf{x}, \mathbf{e}) = \mathbf{w}_{j,l,t} \quad (15)$$

Equations (13)–(15) constitute the moment conditions to be tested. Equation (13) corresponds to the CD-rationalizability first-order conditions; equation (15) enters the centered measurement error assumption; and equation (14) is a user-defined moment recovering the effects of interest, as shown by Aguiar and Kashaev (2020).

Since prices are not simulated with measurement error, observed prices equal actual prices and the measurement error vector contains exclusively consumption errors. In equation (14), since $\mathbf{a}_{l,t} \in [0, 1]$, we have $\theta \in [-2, 2]$. The baseline slack variable’s value for θ is taken to be the unbiased DiD estimator. For a sufficiently broad testing interval, a set of values for θ can be found such that the model is not rejected. The

full moment conditions corresponding to equations (13)–(15) are:

$$\mathbb{E}[g_{A,j,l,t}(\mathbf{x}, \mathbf{e})] = 0 \quad (16)$$

$$\mathbb{E}[g_{D,j,l=1}(\mathbf{e}; \theta)] = 0 \quad (17)$$

$$\mathbb{E}[g_{M,j,l,t}(\mathbf{x}, \mathbf{e})] = 0 \quad (18)$$

Before introducing the econometric framework, consider the following synthesis of equations (16) and (17):

$$g_{I,j,l,t}(\mathbf{x}, \mathbf{e}; \theta) = (g_{A,j,l,t}(\mathbf{x}, \mathbf{e})', g_{D,j,l=1}(\mathbf{e}; \theta)') \quad (19)$$

$$g_{j,l,t}(\mathbf{x}, \mathbf{e}; \theta) = (g_{A,j,l,t}(\mathbf{x}, \mathbf{e})', g_{D,j,l=1}(\mathbf{e}; \theta)', g_{M,j,l,t}(\mathbf{x}, \mathbf{e})') \quad (20)$$

The conditions associated with moment equations (16)–(18) therefore reduce to:

$$\mathbb{E}[g_{j,l,t}(\mathbf{x}, \mathbf{e}; \theta)] = 0 \quad (21)$$

where $\mathbf{x} = (\mathbf{c}_{j,l,t}, \mathbf{p}_{j,l,t})$ and $\mathbf{e} = (\mathbf{w}_{j,l,t}, \mathbf{a}_{j,l,t})$. For each household, there are k moment functions corresponding to equation (16) and one additional function from equation (17), plus q moment functions corresponding to the centered measurement error condition in (18). Testing (21) allows for consistent inference of the estimated parameter $\hat{\beta}$ while accounting for unobserved measurement error.

The left-hand side of (21) can be written as:

$$\mathbb{E}_{\mu \times \pi}[g_{j,l,t}(\mathbf{x}, \mathbf{e}; \theta)] = \int_X \int_{E|X} g(x, e; \theta) d\mu d\pi \quad (22)$$

where $\mu \in \mathcal{P}_{E|X}$ and $\pi \in \mathcal{P}_X$. As shown by Aguiar and Kashaev (2020), for a random array $(\mathbf{p}_t, \mathbf{c}_t)_{t \in \mathcal{T}}$ to be approximately s/CD -rationalizable such that Assumption 1 holds is equivalent to:

$$\inf_{\mu \in \mathcal{P}_{E|X}} \|\mathbb{E}_{\mu \times \pi}[g(\mathbf{x}, \mathbf{e}; \theta)]\| = 0 \quad (23)$$

where $\pi_0 \in \mathcal{P}_X$ is the observed distribution of $\mathbf{x}$. This is a preliminary approach to our testing conditions, which amounts to testing, at a given significance level, whether there exists an unknown distribution of latent variables conditional on observables such that the moment conditions are fulfilled.

To deal with the test of an unknown distribution of latent vectors, we apply the Entropic Latent Variable Integration via Simulation (ELVIS) approach by Schennach (2014), also used in Aguiar and Kashaev (2020). This approach allows us to write our testing conditions in terms of the observable distributions only.

Following the notation of Aguiar and Kashaev (2020), the maximum-entropy (ME) moment of $g(x, e; \theta)$ can be written as:

$$h(x; \gamma, \theta) = \int_{e \in E|X} g(x, e; \theta) d\eta^*(e|x; \gamma, \theta) \quad (24)$$

where

$$\left\{ d\eta^*(\cdot|x; \gamma, \theta) = \frac{\exp(\gamma' g(x, \cdot)) d\eta(\cdot|x)}{\int_{e \in E|X} \exp(\gamma' g(x, e; \theta)) d\eta(e|x)}, \quad \gamma \in \mathbb{R}^{k+q} \right\} \quad (25)$$

is a family of exponential conditional probability measures. Accordingly, the ME moment of $g(x, e; \theta)$ is:

$$h(x; \gamma, \theta) = \frac{\int_{e \in E|X} g(x, e; \theta) \exp(\gamma' g(x, e; \theta)) d\eta(e|x)}{\int_{e \in E|X} \exp(\gamma' g(x, e; \theta)) d\eta(e|x)} \quad (26)$$

where $\gamma \in \mathbb{R}^{k+q}$ is a nuisance parameter and $\eta \in \mathcal{P}_{E|X}$ is an arbitrary user-input distribution supported on $E|X$ such that

$$\mathbb{E}_{\pi_0} [\log \mathbb{E}_{\eta} [\exp(\gamma' g(\mathbf{x}, \mathbf{e}; \theta)) | \mathbf{x}]] \quad (27)$$

exists and is twice continuously differentiable in γ .

Thus h is the marginal moment of g where the latent vector has been “integrated out” using a probability measure from the family (25). An important implication follows from Aguiar and Kashaev (2020): for a random array $(\mathbf{p}_t, \mathbf{c}_t)_{t \in \mathcal{T}}$ to be approximately s/CD-rationalizable such that Assumption 1 holds is equivalent to:

$$\inf_{\gamma \in \mathbb{R}^{k+q}} \|\mathbb{E}_{\pi_0} [h(\mathbf{x}; \gamma, \theta)]\| = 0 \quad (28)$$

Importantly, the choice of η does not affect (28), only the nuisance parameter γ .

This is a necessary and sufficient condition: the existence of any set of distributions in the family (25) such that the ME moment property in (26) holds implies the existence of some set of distributions, conditional on unobserved latent variables, such that condition (23) is fulfilled. More importantly, this implies that $\mathbf{x}$ is rationalized by the stated preference structure, fulfills Assumption 1, and that our model of interest holds for some nonempty subset $\theta \in \Theta$ —all testable without relying on distributions conditional on unobserved latent variables.

4.1. Implementation

From (28), the following condition applies: for a random array $(\mathbf{p}_t, \mathbf{c}_t)_{t \in \mathcal{T}}$ to be approximately consistent with s/CD-rationalizability such that Assumption 1 holds, is equivalent to:

$$\min_{\gamma_M \in \mathbb{R}^q} \|\mathbb{E}_{\pi_0} [\tilde{h}_M(\mathbf{x}; \gamma, \theta)]\| = 0 \quad (29)$$

with:

$$\tilde{h}_M(x; \gamma, \theta) = \frac{\int_{e \in E|X} g_M(x, e) \exp(\gamma' g_M(x, e)) \mathbb{I}(g_I(x, e; \theta) = 0) d\eta(e|x)}{\int_{e \in E|X} \exp(\gamma' g(x, e; \theta)) \mathbb{I}(g_I(x, e; \theta) = 0) d\eta(e|x)} \quad (30)$$

This is proved by Aguiar and Kashaev (2020), who also show that for any divergent sequence $\{\gamma_{I,l}\}_{l=1}^{+\infty}$,

approximate s/CD-rationalizability is equivalent to the sequence $\{\gamma_{M,l}\}$ of minimizers of

$$\lim_{l \rightarrow +\infty} \min_{\gamma_M \in \mathbb{R}^q} \left\| \mathbb{E}_0[h(\mathbf{x}; (\gamma'_{M,l}, \gamma'_M), \theta)] \right\| = 0 \quad (31)$$

being convergent to some finite γ_0 not depending on $\{\gamma_{M,l}\}_{l=1}^{+\infty}$. The optimization procedure therefore depends only on $\gamma_M \in \mathbb{R}^q$, not on $\gamma \in \mathbb{R}^{k+q}$. Given the DGP calibration, $q = (T \times K \times \tau c) + 1 = 13$, where τc is the number of interest groups (treatment and control). In practice, the integrals in (30) are computed using Markov Chain Monte Carlo methods—specifically, a Metropolis-Hastings algorithm sampling from η .

For a given observed data set $\{\mathbf{x}_i\}_{i=1}^n = \{(\mathbf{p}_{t,i}, \mathbf{c}_{i,t})_{t \in \mathcal{T}}\}_{i=1}^n$ with sample size $n \in \mathbb{N}$, the finite-sample analogues of the ME moment condition and its corresponding variance are:

$$\hat{h}_M(\gamma, \theta) = \frac{1}{n} \sum_{i=1}^n \tilde{h}_M(\mathbf{x}_i; \gamma, \theta) \quad (32)$$

$$\hat{\Omega}(\gamma, \theta) = \frac{1}{n} \sum_{i=1}^n \tilde{h}_M(\mathbf{x}_i, \gamma, \theta) \tilde{h}_M(\mathbf{x}_i, \gamma, \theta)' - \hat{h}_M(\gamma, \theta) \hat{h}_M(\gamma, \theta)' \quad (33)$$

The test statistic proposed by Aguiar and Kashaev (2020), derived from Schennach (2014), is then:

$$TS_n = n \inf_{\gamma \in \mathbb{R}^q} \hat{h}_M(\gamma, \theta)' \hat{\Omega}(\gamma, \theta)^{-1} \hat{h}_M(\gamma, \theta) \quad (34)$$

Under the null hypothesis that the data is approximately consistent with s/CD-rationalizability:

$$\lim_{n \rightarrow \infty} \mathbb{P}(TS_n > \chi_{q,1-\alpha}^2) \leq \alpha \quad (35)$$

for any $\alpha \in (0, 1)$. The confidence set for θ_0 is obtained by inverting $TS_n(\theta_0)$:

$$\left\{ \theta_0 \in \Theta : TS_n \leq \chi_{13,1-\alpha}^2 \right\} \quad (36)$$

There is no need to test rationalizability independently prior to the model's identified set: rejection of rationalizability would imply an asymptotically empty $\theta_0 \in \Theta$. In our case, the confidence set of the structural DiD model in (36) is tested simultaneously with measurement error robustness and Cobb-Douglas rationalizability (which is controlled by our DGP).

5. Results

Table 1 summarizes the reduced-form DiD estimation for two different simulation settings. In both cases, the estimation with observed data yields a significance rejection rate above 11%, compared to 0% when estimated with real data. Additionally, the misinference rate—the frequency with which the estimated parameter falls outside the real-data confidence interval—exceeds 45% in both cases. Interestingly, the estimated effect with observed data falls below the estimated effect with real data in both cases, a bias attributable to measurement

Table 1. Reduced-form DiD estimation performance with and without measurement error

	1,000 Repetitions		100 Repetitions	
	Observed Data	Real Data	Observed Data	Real Data
Estimated effect	0.1089	0.1242	0.1068	0.1247
Standard error	0.0344	0.0122	0.0349	0.0128
Confidence interval	(0.0424, 0.1753)	(0.1001, 0.1483)	(0.0407, 0.1729)	(0.1007, 0.1488)
Significance rejection rate	0.112	0.00	0.130	0.00
Significance level		95%		95%
Misinference rate	0.495		0.460	

Notes: Reduced-form DiD estimation for both observed data (suffering from measurement error) and real data, according to the simulated DGP. Results are averages over 1,000 and 100 simulation repetitions respectively. The misinference rate denotes the share of repetitions in which the observed-data estimated effect falls outside the real-data estimator’s 95% confidence interval.

error. Figures 1 and 2 illustrate the direction and magnitude of this bias at the confidence interval level.

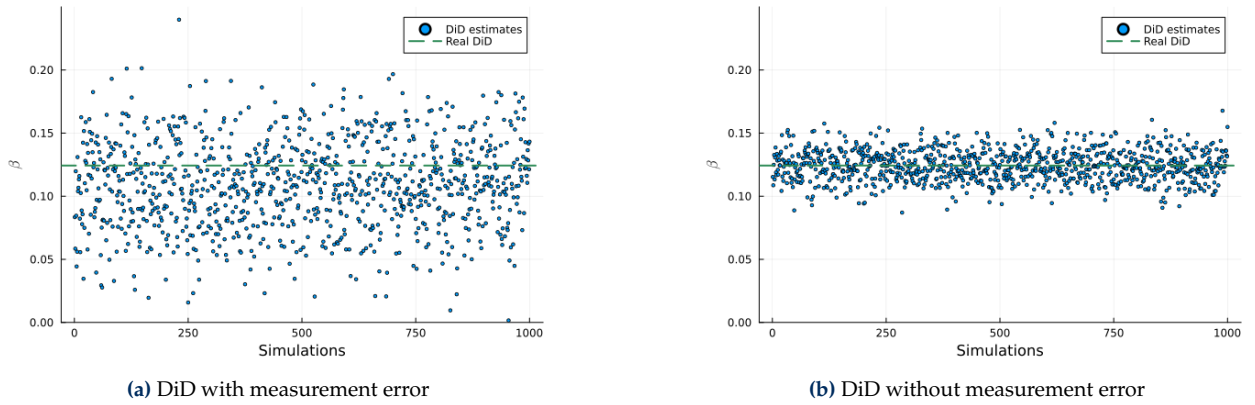
**Figure 1.** Difference-in-Differences estimates across 100 simulations

Table 2 reports two simulations of the structural estimation model described in Section 4. In both cases, a grid of size 10 was used over the ATT slack parameter θ . Simulation 1 was estimated taking the reduced-form confidence interval (real data) as bounds. While approximately 87–88.8% of the time the reduced-form estimation with observed data is not rejected (as shown in Table 1), the structural model, by accounting for measurement error, yields a more precise non-rejection region. Simulation 2 ran over a subset of the original grid where the test statistic achieved parabolic convergence to a deeper non-rejection region.

Recall that the model in Section 4 is constructed with a structural significance parameter $\alpha = 0.05$. We provide Monte Carlo evidence that this parameter is identified over a sharp set in $(0.1294, 0.1359)$ at the 0.1 level. Since we are interested in set identification rather than point identification, the slight upward shift of this set relative to the causal effect point is attributable to the relatively small simulation sample. Stronger asymptotic consistency would be attained by running 1,000 rather than 100 repetitions per θ -value—an extension not implemented due to computational cost.

6. Discussion

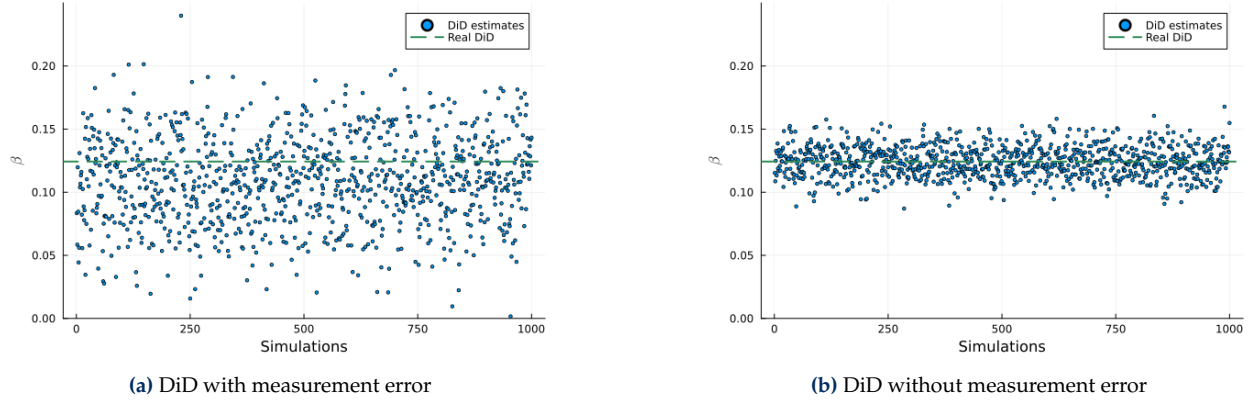


Figure 2. Difference-in-Differences estimates across 1,000 simulations

Table 2. Structural DiD set-identified estimation of the ATT slack parameter

Simulation 1: Extended Grid		Simulation 2: Reduced Grid	
θ	Rejection rate	θ	Rejection rate
0.0424	1.00	0.1162	0.30
0.0572	1.00	0.1195	0.18
0.0719	1.00	0.1228	0.13
0.0867	0.96	0.1261	0.12
0.1015	0.73	0.1294	0.10
0.1162	0.30	0.1326	0.10
0.1310	0.10	0.1359	0.09
0.1458	0.27	0.1392	0.12
0.1605	0.73	0.1425	0.18
0.1753	0.95	0.1458	0.27

Notes: Structural DiD set-identified estimation of the slack parameter for the ATT with observed data, across two simulation tranches with 100 repetitions each. Simulation 1 uses bounds equal to the confidence interval from the reduced-form DiD with observed data. Simulation 2 refines the grid to a subset of Simulation 1 where convergence to high non-rejection was achieved. In both simulations, the average real effect was $ATT = 0.1237$.

Under a controlled simulated-data environment, it has been shown that observed data sets suffering from measurement error yield biased causal DiD estimations. For the proposed DGP calibration, this bias can be calculated in sign and magnitude as shown in Table 1. An alternative causal estimation method is proposed, exploiting the structural solution provided by the stochastic revealed preference approach of Aguiar and Kashaev (2020) to account for measurement error. This methodology applies a Maximum-Entropy Moment Conditions structure derived from the ELVIS methodology of Schennach (2014). Monte Carlo evidence demonstrates that this method provides set-identification robust to classical forms of measurement error when estimating the causal effect of an exogenous shock on consumption bundles. The identified set of the interest parameter suggests that, when relying on observational data with unobserved measurement error, conventional reduced-form DiD techniques may yield biased estimators and wide confidence intervals, only a small subset of which contains an identification region that properly accounts for unobserved randomness.

Several important extensions remain for future work. First, stronger computational capacity is needed and the implementation mechanism should be optimized so that the Monte Carlo estimation of the structural

model yields asymptotic consistency at the 0.05 level for each θ in the grid, and optimization methods in the algorithm could be further refined. Second, the identified set over which the structural estimated effect achieves non-rejection is potentially consistent with a subset of distributions satisfying parallel-trends conditions; however, this assumption should be explicitly modeled as an additional moment condition in equations (13)–(15) to narrow the set to a space of pure causal effects.

Third, while CD preferences cover a wide range of stochastically diverse consumer behavior types, a robustness extension would consist of expanding the model to other flexible preference structures—such as CES—making parameters empirically interpretable in a revealed preference context, particularly given that parametric rationalizability is implicitly tested in our moment conditions. Finally, while the use of classical measurement error has been justified here, it would be desirable to consider alternative types of unobserved error. Other forms of non-classical systematic measurement error lead to more severe biases for which the applied methodology would remain robust. These constitute promising directions for future extensions of this paper.

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Appendices

A. Proof of Theorem 1

Proof. We first show that CD-rationalizability is equivalent to the existence of some matrix $(\alpha_{l,t})_{l \in \mathcal{L}, t \in \mathcal{T}}$ with $\sum_l \alpha_{l,t} = 1$ and $c_{l,t} = \alpha_{l,t} y_t / p_{l,t}$ for all $l \in \mathcal{L}, t \in \mathcal{T}$.

Consider the convexity-preserving monotone transform of equation (1): $\ln u = \sum_l \alpha_l \ln c_l$. It will be shown that there exists $(\alpha_{l,t})$ such that

$$c_l = \arg \max_{c_l} \left\{ \sum_l \alpha_l \ln c_l \text{ s.t. } p'c = y \right\}. \quad (37)$$

Solving the first-order conditions of the Lagrangian

$$\mathcal{L} = \sum_l \alpha_l \ln c_l + \lambda [y - p'c] \quad (38)$$

yields the marginal condition

$$\alpha_l = \lambda p_l c_l. \quad (39)$$

Since $\ln u$ is concave and the budget set $p'c = y$ is convex, this system is perfectly identified with a unique interior solution. Summing (39) over $k \in \mathcal{L}$ gives $\lambda = \sum_k \alpha_k / \sum_k p_k c_k$, which upon substitution yields:

$$c_l = \frac{\alpha_l}{p_l \sum_k \alpha_k} \sum_k p_k c_k. \quad (40)$$

Noting that $\sum_l p_l c_l \equiv p'c$ and using $\sum_l \alpha_l = 1$, the demand for good $l \in \mathcal{L}$ simplifies to:

$$c_l = \frac{\alpha_l y}{p_l} \quad (41)$$

for each $t \in \mathcal{T}$ and any $j \in \mathcal{J}$.

Uniqueness. Suppose there exists another consumption level $\tilde{c}$ such that $u(\tilde{c}) > u(c)$ and $p'\tilde{c} = y$. This contradicts the fact that c satisfies the first-order conditions of the strictly concave Lagrangian. Furthermore, if $p'\tilde{c} > y$, this solution lies outside the convex budget set and is therefore infeasible. Thus c as in (41) is the unique solution.

This implies that for a demand $c_{l,t} = \alpha_{l,t} y_t / p_{l,t}$ to exist with its matrix $(\alpha_{l,t})_{l \in \mathcal{L}, t \in \mathcal{T}}$ is equivalent to stating that $(c_t)_{t \in \mathcal{T}}$ solves (37) and therefore $(p_t, c_t)_{t \in \mathcal{T}}$ is CD-rationalizable. $\square$

B. Data Generating Process

The data is generated by the following algorithm, which outputs the observed data set of consumption and prices.

Step 0.

set $\mathcal{T} = \{1, 2\}$ and $\mathcal{L} = \{1, 2, 3\}$ for $\mathcal{J} = \{1, \dots, 200\}$ households.
set control and treatment groups $\mathcal{C}, \tau \subset \mathcal{J}$ by randomly drawing equal groups, $|\mathcal{C}| = |\tau| = 100$, with $\tau \cap \mathcal{C} = \emptyset, \tau \cup \mathcal{C} = \mathcal{J}$.
set $(\mathbf{a}_{j,l,t}) \sim \mathcal{N}(\mu = 1/K, \sigma^2 = 0.05)$ truncated at $[0.1, 0.9]$ for the simulated population.
set $\mathbf{a}_{j,1,t} \in (0, 1)$ and $\sum_l \mathbf{a}_{j,l,t} = 1$ for all $t \in \mathcal{T}, j \in \mathcal{J}$.
set $\mathbf{a}_{j,l=1,t}$ as receiving a positive exogenous scaled shock from $\psi_{j \in \tau} \sim \text{Unif}(0, 1)$ for the treatment group.
set observed prices $(\mathbf{p}_{l,t}) \sim \text{Exp}(\lambda = 1)$.
set income $(y_{j,t}) = 1$ for all $t \in \mathcal{T}, j \in \mathcal{J}$.
set demand $\mathbf{c}_{j,l,t} = \mathbf{a}_{j,1,t} y_{j,t} / \mathbf{p}_{l,t}$.
set measurement error $\epsilon_{j,l,t} = (b - a) \times \zeta_{j,l,t} + a$ with $\zeta_{j,l,t} \sim \text{Unif}(0, 1), b = 1.9, a = 0.1$.
set observed demand $\mathbf{c}_{j,l,t}^* = \mathbf{c}_{j,l,t} \times \epsilon_{j,l,t}$.
return observed data set $\mathbf{x} = \{(\mathbf{c}_{j,l,t}^*, \mathbf{p}_{l,t})\}_{j,l,t}$.

C. Algorithm for Section 4

Step 0.

set a realization of $\{(\mathbf{c}_{j,l,t}^*, \mathbf{p}_{l,t})\}_{j,l,t}$.
fix a user-defined grid $\Theta = \{\theta_k\}_{k=1}^{\Theta_s}$.
fix $g_I = (g'_A, g'_D)$ and g_M .
fix $\eta \in \mathcal{P}_{E|X}$.
fix $\mathbf{x} = (\mathbf{x}_i)_{i=1, \dots, n}$ where $\mathbf{x}_i = \{(\mathbf{c}_{t,i}^*, \mathbf{p}_{t,i})\}_{t \in \mathcal{T}}$ is the i -th observation.

Step 1 (Integration).

set $i = 1$; **fix** cl (MCMC chain length), $\eta, \gamma, \mathbf{x}_i$, and initial chain element $e_{-\text{nburn}}$ satisfying constraints.
set $r = -\text{nburn} + 1$ and $\hat{h}_M(x_i, \gamma) = 0$.
while $r \leq \text{nsims}$ **do**
 draw $e_{\text{jump}} = \{(\alpha_{j,l,t}), w^c\}$ proportional to $\tilde{\eta}(\cdot | x_i, \theta) = \eta(\cdot | x_i) \mathbb{I}(g_I(x_i, \theta, \cdot) = 0)$
 draw $\alpha \sim \text{Unif}(0, 1)$
 set $e_r = e_{\text{jump}}$ if $[g_M(x_i, e_{\text{jump}}) - g_M(x_i, e_{r-1})]' \gamma > \log(\alpha)$
 if $r > 0$ **then**
 $\tilde{h}_M(x_i, \gamma, \theta) += g_M(x_i, e_r) / cl$
 end if
 $r += 1$
end while
set $i = 1$

while $i \leq n$ **do**

define $\tilde{\eta}(\cdot|x_i, \theta) = \eta(\cdot|x_i) \mathbb{I}(g_I(x_i, \theta, \cdot) = 0)$

integrate latent variables by $\tilde{\eta}(\cdot|x_i, \theta)$ to obtain $\tilde{h}_M(x_i, \gamma, \theta)$

$i += 1$

end while

compute $\hat{h}_M(\gamma, \theta) = \frac{1}{n} \sum_i \tilde{h}_M(\mathbf{x}_i, \gamma, \theta)$ and

$$\hat{\Omega}(\gamma, \theta) = \frac{1}{n} \sum_i \tilde{h}_M(\mathbf{x}_i, \gamma, \theta) \tilde{h}_M(\mathbf{x}_i, \gamma, \theta)' - \hat{h}_M(\gamma, \theta) \hat{h}_M(\gamma, \theta)'$$

compute objective function $\text{ObjFun}(\gamma, \theta) = n \times \hat{h}_M(\gamma, \theta)' \hat{\Omega}(\gamma, \theta) - \hat{h}_M(\gamma, \theta)$

Step 2 (Optimization).

compute

$$TS_n = \min_{\gamma} \text{ObjFun}(\gamma, \theta)$$